

LAB REPORT- LAB 1- 3/2/25
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CS23B096

Q1) Please find the array contents at the beginning. The array is [5,3,4,6,5]

```
(spike) mem 0 0x00000000000011468
0x0000000000000005
(spike) mem 0 0x00000000000011470
0x0000000000000003
(spike) mem 0 0x00000000000011478
0x0000000000000004
(spike) mem 0 0x00000000000011480
0x0000000000000006
(spike) mem 0 0x00000000000011488
0x0000000000000005
```

Please find the array contents at the end of the iterations. The final array should be [5,4,6,9,9].
Please make note of the spike bug where at alternate occurrences only one word appears.

```
(spike) until pc 0 10150
bbl loader
(spike) until pc 0 10170
(spike) mem 0 0x11468
0x0000000400000005
(spike) mem 0 0x1146c
0x0000000000000004
(spike) mem 0 0x11470
0x0000000900000006
(spike) mem 0 0x11474
0x0000000000000009
(spike) mem 0 0x11478
0x0000000000000009
(spike) □
```

After 2 iterations-

Array should be [5,4,4,6,5]

```

I (spike) mem 0 0x11468
0x0000000400000005
(spike) mem 0 0x1146c
0x0000000000000004
(spike) mem 0 0x11470
0x0000000600000004
(spike) mem 0 0x11474
0x0000000000000006
(spike) mem 0 0x11478
0x0000000000000005

```

Report for Q1- I define 5 numbers and store them in variables named num1,num2,num3,num4,num5.

I first store the address of num1 in t0, and t1 and t6 store numbers 0 and 5. t1 is the iterator and t6 is to check the breaking condition of the loop.

Loop functions as follows-

t5 will store the array number under consideration

Value from t2 is updated back to t0

Incrementing t0 by 4 for the next array element, incrementing t1 by 1 for the iterator

COMMANDS USED: mem for obtaining the value stored at an address and reg for register address

Q2) The numbers num1 and num2 are 6 and 4 respectively.

```

(spike) reg 0 t0
0x0000000000011498
(spike) mem 0 0x0000000000011498
0x0000000000000006
(spike) reg 0 t1
0x00000000000114a0
(spike) mem 0 0x00000000000114a0
0x0000000000000004
(spike) □

```

Please find addition below

```

core 0: 0x00000000000010164 (0x01c38eb3) add    t4, t2, t3
(spike)
core 0: 0x00000000000010168 (0x81018f13) addi    t5, gp, -2032
(spike)
core 0: 0x0000000000001016c (0x01df2023) sw      t4, 0(t5)
(spike) reg 0 t5
0x000000000000114a8
(spike) mem 0 0x000000000000114a8
0x0000000000000000a

```

Please find subtraction below-

```

core 0: 0x00000000000010170 (0x41c38eb3) sub    t4, t2, t3
(spike)
core 0: 0x00000000000010174 (0x81818f13) addi    t5, gp, -2024
(spike)
core 0: 0x00000000000010178 (0x01df2023) sw      t4, 0(t5)
(spike) reg 0 t5
0x000000000000114b0
(spike) mem 0 0x000000000000114b0
0x00000000000000002

```

Please find AND below-

```

core 0: 0x0000000000001017c (0x01c3feb3) and    t4, t2, t3
(spike)
core 0: 0x00000000000010180 (0x82018f13) addi    t5, gp, -2016
(spike)
core 0: 0x00000000000010184 (0x01df2023) sw      t4, 0(t5)
(spike) reg 0 t5
0x000000000000114b8
(spike) mem 0 0x000000000000114b8
0x00000000000000004

```

Please find OR below-

```

core 0: 0x00000000000010188 (0x01c3eeb3) or      t4, t2, t3
(spike)
core 0: 0x0000000000001018c (0x82818f13) addi    t5, gp, -2008
(spike)
core 0: 0x00000000000010190 (0x01df2023) sw      t4, 0(t5)
(spike) reg 0 t5
0x000000000000114c0
(spike) mem 0 0x000000000000114c0
0x0000000000000006

```

Please XOR below-

```

core 0: 0x00000000000010194 (0x01c3ceb3) xor      t4, t2, t3
(spike)
core 0: 0x00000000000010198 (0x83018f13) addi    t5, gp, -2000
(spike)
core 0: 0x0000000000001019c (0x01df2023) sw      t4, 0(t5)
(spike) reg 0 t5
0x000000000000114c8
(spike) mem 0 0x000000000000114c8
0x0000000000000002

```

Report for Q2-

Num1 and num2 are 6 and 4 respectively.res1,...,res5 are for the results

t0 and t1 store the addresses of num1 and num2 respectively.

We iteratively do the following-

Perform the instruction

Load address of res variable to t5

Store value of t4 to t5

Hence, res1 gets addition value,, res5 gets XOR

COMMANDS USED: mem for obtaining the value stored at an address and reg for register address

